

Honors Calculus III, Homework 6

Please upload solutions to **Questions 1(c), 2, 3 and Problem 2-7** by **8pm on Monday, May 08**. You can use any results you want from lectures, provided you state them clearly, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says “optional”. As usual, for full credit you need to show and explain your work.

- For each of the following sets, determine with proof whether they are open, closed, compact or connected.
 - $B(0, 1) \subset \mathbf{R}^n$.
 - $\overline{B}(0, 1) \subset \mathbf{R}^n$.
 - (5pt) $\{(x, y) \in \mathbf{R}^2 : (x + y - 4)(x^2 + y^2 - 1) = 0\}$.
 - $\{x \in \mathbf{R}^n : \max_i \{|x_i|\} \leq 1\}$.
 - $\{(x, y) \in \mathbf{R}^2 : y = \sin(x)\}$.
- (5pt) Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be a linear map. Prove that there exists $y \in \mathbf{R}^n$ such that $f(x) = \langle x, y \rangle$ for all $x \in \mathbf{R}^n$.
- (5pt) Let $U = \{(x, y) \in \mathbf{R}^2 : y \neq 0\}$. Prove from the definition that the function

$$f : U \rightarrow \mathbf{R}, f(x, y) = x/y$$

is differentiable at $(1, 1)$ with derivative

$$Df(1, 1)(h, k) = h - k.$$

- Problems from Spivak, *Calculus on manifolds*: 2-2, 2-7(5pt), 2-8, 2-17, 2-20.